

ACTIVITY 19 — FACILITATOR GUIDE**Kanban Board with WIP Limits and Lead/Cycle Time**

TYPE Analysis	DURATION 45 minutes	MAPS TO LO5	TOPIC 5 - Monitor and Control the Project
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Purpose of this activity

By the end of this activity learners will be able to build a Kanban board that reflects Contoso's actual workflow rather than a generic three-column template.

What learners produce

kanban board, flow metrics, bottleneck analysis — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- The Project Management Templates pack (see the Activities folder)
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Design the workflow columns. A three-column To Do / Doing / Done board hides everything interesting. Contoso's real workflow has distinct stages, and the waiting states between them are where the delay actually lives.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
2	Set WIP limits. Open Kanban Board and create the columns above with these WIP limits.	If a group finishes early, ask what changes at twice the scale or half the budget.
3	Load the real card data. These are the 18 stories completed by the Contoso team during sprints 8 and 9, with the dates each card entered the system, started development, and was accepted. All dates are working	Ask for one item the group nearly dropped and what made them keep it.

	days measured from the start of sprint 8 as day 1.	
4	Compute the flow metrics	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
5	Apply Little's Law	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
6	Find the bottleneck. A bottleneck reveals itself as a queue that grows in front of one stage. Here is the average number of cards in each column, sampled daily across the 20 days.	Ask for one item the group nearly dropped and what made them keep it.
7	Read the cumulative flow diagram. The Kanban tool will render a cumulative flow diagram. Learn to read it, because a CFD is a strong candidate for a graphic-based exam item.	Watch for groups listing only the obvious. Ask what they are deliberately leaving out, and why.
8	Answer the exam-style scenarios	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.

Suggested timing

Minutes	Phase
4 min	Brief the scenario, confirm the inputs and assign roles
17 min	Work the first half of the steps individually, then pool as a group
17 min	Complete the remaining steps and finalise the deliverable
7 min	Group share-back, facilitator debrief and exam-style linkage

Total: 45 minutes

Discussion & decision prompts

1. Build a Kanban board that reflects Contoso's actual workflow rather than a generic three-column template — how did your group decide, and what would change that decision?
2. Set and justify WIP limits, and understand why limiting work in progress speeds delivery — how did your group decide, and what would change that decision?
3. Measure lead time, cycle time and throughput from real card data — how did your group decide, and what would change that decision?
4. Your board separates active columns from queue columns. A board with only In Progress and Done cannot show you where the waiting happens — can every member of your group defend it?
5. Where would kanban board with wip limits and lead/cycle time have saved you time or money on a real project you have worked on?

What good looks like

- Your board separates active columns from queue columns. A board with only In Progress and Done cannot show you where the waiting happens.
- Your mean lead time is 13.11 days and your mean cycle time is 6.94 days, and you can state why they differ.
- Your flow efficiency is 52.9%, and you interpreted it as "half the customer's wait is pure queue" rather than as a performance grade.
- Your Little's Law check produced an implied WIP of about 6.25 against an observed 6, and you used that agreement to justify using the law predictively.

- You showed arithmetically that doubling WIP roughly doubles cycle time while delivering nothing sooner.
- You identified Test as the bottleneck and cited its 100% utilisation, the 97% queue in front of it, and the collapse to 33% utilisation downstream.
- You explicitly rejected raising the In Test WIP limit and explained why it would make cycle time worse.
- Your constraint response includes subordination - developers stopping new work to help clear test - and you recognised that developers appearing less busy is the correct outcome rather than a problem.